



OUTDOOR INFRARED SAUNA BUYING GUIDE



Can You Set Up An Infrared Sauna Outdoors?

After a long winter cooped up indoors, people are longing to get outside and soak up a little Vitamin D. They, however, are wondering if they can soak up the far infrared (IR) rays of their infrared saunas outdoors.

And who could blame them?

The answer is Yes, you can set up your infrared sauna outside, but there are some things you should consider first, namely—the warranty, the power supply, and the effect the elements will have on your sauna.

But having set up many infrared saunas outside, we have some tips for you. Let's go!

Check your warranty

Most infrared saunas come with some sort of warranty, and many come with a lifetime warranty. Check the fine print of your warranty, however—most have a clause for outdoor storage that will void it if stored outside.

There are ways around this, too. For instance, at Outdoor Infrared Saunas®, if you want to keep your infrared sauna outside you are totally covered for a period of 3 years. If

you want to add full lifetime coverage, simply pay an additional 20 percent upon checkout of your order.

Bottom line—Check the details of your warranty or call customer service before setting up your sauna outside to make sure you're on the same page with your company's outdoor warranty.

The effects of sunlight on your infrared sauna

Believe it or not, the UV rays of the sun are usually more harsh to the exterior wood of the sauna than anything else. Long-term exposure to sunlight can affect the look of the exterior wood. It won't necessarily harm the functional aspects of your sauna, but can have a negative effect on the appearance.

Use this knowledge in terms of where you will put your sauna outside. It might be better under a covered patio instead of in direct sunlight.

Bottom line—Different wood reacts differently to exposure to sunlight.

Be sure to check out the next chapter: Red Cedar, Spruce, or Poplar? What Is the Best Type of Wood for an Infrared Sauna?

The effects of rain and snow on your infrared sauna

Snow in the Spring? Depending on where you live, the elements can be harsh on the wood of your infrared sauna.

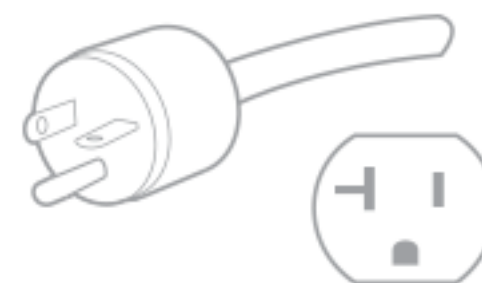
While it's true that most infrared saunas are designed to withstand some moisture, if pools of water are left standing on top of or around the base of your sauna, you may see some negative effects like mold or warped wood. Usually this won't affect the actual operation of the sauna, but you'll need to check with your model.

Bottom line—Keep the rain and snow off with a cover to stop the moisture from lying directly on top of the wood for extended periods of time.

At Outdoor Infrared Saunas® we only use sloped and peaked roofs to make sure all snow and rain don't pool on your roof.

Power supply

Again, we can't speak for all infrared saunas, but every model of Outdoor Infrared Saunas® requires at least a 120 volt plug. I used to think that 110 volt and 120 volt plugs were the same (whoops!), but they're not. 110-volt plugs have the two vertical prongs and 120-volt plugs have three prongs (with one horizontal and one vertical). Like this:



It's pretty easy to install a 120-volt outlet, but most homes in the United States don't come standard with them outdoors, so you'll need to account for that.

Also, you will need a dedicated 20 amp circuit for your sauna—most homes only offer 15 amp circuits standard, so that will be something to think about as well.

The plug is shorter for outdoor saunas, as well. Because of this you'll need your outlet within 6 feet of your sauna.

Bottom line—Make sure you have the right plugs in place for your saunas power requirements.

Space planning

We can't speak for every type of infrared sauna, but most of our saunas weigh between 500-1000 lbs and take up at least 4' x 4'. Make sure that the surface you're going to put your sauna on can handle the weight.

For detailed specifications on all our different sauna models, including drawings, check out our product pages at www.outdoorinfraredsauna.com

Buy a sauna specifically designed for outdoor use!

Yes, we saved the best for last. If you're keen on permanently storing your infrared sauna, why not buy one that's specifically designed to stand up to the elements?

Our newest line of IR saunas, the outdoor Rustic and Sierra lines, are purpose-built for the outdoors. The main differences between our traditional line of saunas are:

- Heavy duty cedar wood with insulated cabins to make sure the outside of your sauna does not leak or fade. Nor will it leak heat.
- 1000 pound sauna that will not be moved or destroyed by the outside elements.
- 2 500 watt full spectrum heaters to make sure your sauna can heat up even in the coldest environments.

If you're interested in learning more about outdoor saunas, please don't hesitate to request a catalogue or get in touch with us.

What Kind of Wood is Best?

When I first became involved in the infrared sauna industry, I assumed the wood that makes up the panels of the cabin and the bench was purely aesthetic. When I'd read sauna specs and see the section on wood, I assumed that people picked different types of wood based mostly on their home decorating scheme. I mean, these saunas are serious pieces of furniture, large boxes of elegant wood that go in the corner of a workout room, or on a deck overlooking a wooded backyard. But, the more I learned about them, the more I discovered that the type of wood used in a sauna is important because every material in a sauna has the potential to affect a user's health.

And that's not the only reason wood matters. The type of wood used makes a difference in how well the sauna will age, how well it will stand up to repeated use, and whether your sauna will be comfortable enough that you'll want to continue using it over time.

Why Wood Matters Within Infrared Saunas

Let's examine, in depth, the biggest factors to consider when choosing a type of wood for your sauna:

- **Wood toxicity:** This is one of the biggest concerns users have when I talk with them about choosing wood for a custom sauna. The majority of potential sauna users come to me because they are interested in detoxifying their bodies. With that in mind, they definitely don't want wood that will introduce new toxins into their environment while they're using the sauna. Toxicity of building material is a concern for any health facility, and infrared saunas are no exception. There is a very simple solution to this problem. Choose a high quality wood like a Western Red Cedar that is naturally antimicrobial and antibacterial, which does not allow for fungus or other allergens to grow. High-quality sauna manufacturers will also specially dry the wood they use in order to burn away oils, resins, and other allergens that may emerge when a sauna is repeatedly heated and cooled during use.

- **Wood weight:** This is a concern for some folks, especially those who are having their sauna delivered over long distances to their homes. Of course, a purchase as significant as a sauna should be made well, durable, and built to last, but harder wood isn't necessarily better. Softer wood won't crack as easily, and it insulates better. Also, wood that is too hard is heavy, and will present logistical problems when it comes time to have it installed in your home. At the same time, the

wood panels in infrared saunas can also warp over time if the wood is not dense enough. The solution to this is to find a wood for your sauna with a density per cubic foot that is on the low side, but not too low. These numbers are typically measured in pounds. In my opinion anything over 30 lb/ft³ will prove to be unnecessarily heavy, but anything lower than 18 lb/ft³ may not hold up to the rigors of use. The neighborhood you want is between 20 and 25 lb/ft³.

- **Resistance to decay:** Warping due to flimsy wood is one serious concern, general decay is another. Wood used in saunas is susceptible to the same problem as any other wood over time: steady decay. While there isn't a metric or numerical system to measure which woods stand up best to decay, there are a few woods that are well known for holding up well over time. The best solution for your sauna is to pick a wood that's naturally decay resistant, and then have any wood you choose treated with a wood preservative to enhance that resistance.

- **Appearance:** This one is pretty much self evident. In my opinion, infrared saunas can look beautiful when you weave them into the decorating scheme in your home or your backyard. This may not be as technical a consideration as other factors, but it's still worth considering. Just take into account the area in which you plan to install your sauna, and ensure that the wood you choose is an aesthetic fit.

- **Crushing strength:** Crushing strength, which is sometimes known as compression strength, is a measurement of how

much weight wood can handle before becoming crushed or broken. Whereas density speaks more to weight, this property speaks more to durability. Your sauna is a haven, a place you can relax and enjoy the many health benefits provided by the process, and that means not having to worry about accidentally cracking the slats of wood in your sauna. Of course, construction and design make a difference too when it comes to durability (that's why you don't want a cheap infrared sauna), but the strength of the actual wood is just as important. I'd recommend against less sturdy woods such as spruce, aspen, and pine.

Types of Wood

While there are too many different types of wood to discuss in depth here, here are some of the kinds of wood that are most commonly used to make saunas:

Western Red Canadian Cedar: Western red cedar has the ideal density for this kind of application, in my opinion, coming in at a spry 23 pounds per cubic foot. It's also a great thermal insulator due to this relatively low density and to its high proportion of air space. It has high durability, and cedar is also naturally antimicrobial and antibacterial, which prevents the growth of fungus or other allergens. Even though infrared saunas are dry (unlike traditional steam saunas), which reduces the likelihood of fungal and bacterial growth, antiseptic qualities are still desirable traits since a sauna is a place where you go to sweat profusely, releasing moisture and bacteria. Cedar is also highly resistant to warping, denting, and

cracking, even given the broad fluctuation of temperatures in saunas. For these reasons, and because of its hardness in wet conditions and its natural ability to repel insects, cedar has been the defacto choice of those who build traditional steam saunas for hundreds of years. These last two qualities also make red cedar perfectly suited for infrared saunas that are going to be kept outdoors.

Basswood: One of the major benefits of basswood is that it is free of harmful toxins and allergens that could bother users; when treated properly it is considered hypoallergenic. Basswood also boasts an excellent strength to weight ratio, and a density that is about 25 pounds per cubic foot.

Poplar: There are many species of poplar, and, unfortunately, most of them are soft and porous, which means they get average marks in tests for strength, resistance to decay, and durability. Poplar isn't all that well suited for varied hot and cold conditions, either, because of its porous nature and middling strength. While it dents easily, it does score great marks for durability, coming in at 22-31 pounds per cubic foot on average.

Spruce: If you've ever done any model building or serious wood carving, you may know this wood, which is as inexpensive as it is low in strength. Simply put, spruce is just too soft to handle the temperature bursts that saunas subject it to. It makes great material for a picture frame, though.

Pine: Honestly, when it comes to wood choices for an infrared sauna, pine of any kind is the worst possible choice, in my

opinion. This wood is low in strength, overly lightweight, and has poor decay resistance. Pine is also more likely to splinter than many of the other choices on our list, and it's high in resin content, which is liable to irritate your eyes and sinuses during use. Stay away.

Pick the Perfect Wood for You

In order to pick the best wood for your sauna, you must know all you can about your options, specifically how each wood interacts with the sauna environment. The section above should have you well-versed by this point, enough to make an educated decision. It's probably clear that I prefer cedar for both indoor and outdoor saunas for all the reasons I've listed above. In addition, I just think it's a good looking wood, and with its red or pinkish hue, it looks sharp in any room, and especially so outdoors on a patio or deck. Nordic spruce is also another good option, with its low-odor, hypoallergenic properties.

With equipment that will affect your personal health, every component matters, from the heaters to the wood panels that make up the sides. There are many health benefits to be gained from regular use of a far infrared sauna, including weight loss, stress relief, and detoxification, but you don't want to limit the potential of your sauna by choosing a wood that prevents it from doing its job properly.

Outdoor Infrared Saunas® only sells sauna made from the highest quality materials— from the heaters to the wood—and built in a way that doesn't cut corners. Over the years, I've

done extensive research into what materials make the best infrared saunas, and I wouldn't sell a sauna that didn't line up with what I've found in my research is best. With Outdoor Infrared Saunas®, you can rest assured that you'll be getting wood of the utmost quality, uniquely suited for the job it's performing.

What Kind of Infrared Heaters are You Getting?

The heater style that you purchase is the single most important part of your infrared sauna buying decision. There are only three styles of infrared sauna heaters: carbon, ceramic and Guru Ultra Carbon®.

Carbon heaters only heat up to 140°F which is not hot enough to raise your core body temperature. Ceramic heaters get up to 400°F which is way too hot to sit 2-3 inches next to. Guru Ultra Carbon® heaters are the most effective at raising your core body temperature at 200°F while still being comfortable enough to sit next to for a 30 minute session.

Guru Ultra Carbon® heaters have the most amount of absorbed infrared. This is because they have the most infrared emissivity combined with the longest infrared wavelength. The second most absorbed is ceramic followed by the weak output of carbon heaters.

Carbon heaters (also known as nano-carbon) are the cheapest infrared heaters to manufacture. This is why you will see them all over the internet for very low prices. However, there is a trade-off: they are riddled with very high Electromagnetic Frequency or EMF. Here is why it is so important to have a low EMF sauna.

Ceramic heaters have much lower EMF on average than carbon heaters. However, you will want to verify a pure ceramic sauna's heaters with a 3rd party EMF test or using your own EMF/gauss meter.

Guru Ultra Carbon® have the lowest EMF levels that have ever been 3rd party tested. This is because they are 2 in 1 heaters with two layers of carbon. By placing two heaters directly on top of each other the EMF is cancelled by negating both the positive and negative currents right on top of each other.

The verdict?

Guru Ultra Carbon® heaters are the most effective heaters at raising core body temperature over a 30 minute period. The carbon elements allow the heaters to produce a long wave infrared heat. This long wave far infrared heat penetrates deeper into the body and is more readily absorbed. The additional carbon compounds gives the heaters a very high and hot infrared output when compared to pure nano-carbon heaters. Because of this we can concentrate 100 percent of the infrared light right on your body.

Heating the air is not the object of infrared therapy, heating your body is.

The concept of carbon panels is nice, but it just doesn't work for an infrared sauna. Companies essentially admit this by their heater size! Think about it, if their infrared heaters were strong enough, wouldn't they want to point 100 percent of the light towards your body? Because of this, they're actually admitting that their panels are too weak to do what an infrared sauna is supposed to do: make you sweat!

One hundred percent of our infrared light energy is pointed at your body—exactly how it should be.

Outdoor Infrared Saunas® uses the best infrared sauna heaters on the market with the best temperature: 200 F°

The single most important component of an infrared sauna is its heater and how hot it gets. No amount of branding, accessories, benches and wood can replace the quality of the infrared heater. Outdoor Infrared Saunas® is the only company that uses a medical grade, clinically-backed Guru Ultra Carbon® combination dual carbon heaters.

Every other infrared sauna uses either carbon fiber or ceramic compound heaters. Both styles have great features—carbon heaters are great for having a lower surface temperature which is closer to the body's own infrared output of 9.4 microns. Ceramic heaters put out a higher amount of infrared at a hotter temperature. However, they both lack what the other has—ceramic gets too hot and has too low of a micron level

(6.0). Carbon is too weak to produce an increase in core body temperature.

In order to attain all of the great health benefits associated with infrared saunas you need to sweat very deeply and sustain that sweat for 30 minutes or more.

Cheap carbon heater's surface temperature of 140 F° will not raise your core temperature high enough to produce a sweat. Ceramic heaters get hot enough to produce a really deep sweat, but don't produce the proper far infrared wavelength in order to be absorbed into your body for deep penetration.

Our Guru Ultra Carbon® heaters offer the best of both carbon and ceramic—the high, hot infrared output of the ceramic with the long, deep penetrating waves of the carbon. Our heaters are much more effective than borderline worthless carbon heaters and they're also much more comfortable to sit three inches away from than ceramic heaters. Remember, if you're not comfortable in your sauna—you won't use it.

Would you like some proof? Our heaters are the only infrared heaters that have been clinically proven to raise core body temperature at the Arcadia Cancer Clinic and to assist in weight loss at Binghamton University.

What Is a Full Spectrum Infrared Sauna? A Guide to Near, Mid and Far Infrared

Far infrared saunas in their current form have been around for 30 years. Near infrared saunas have been around for about half the time as people started realizing the benefits of not just far infrared, but mid and near as well. Let's discuss the merits of having both near, mid and far infrared in a sauna.

The sun's spectrum of electromagnetic radiation has an entire portion that is made up of near, mid and far infrared—which we humans feel as heat. This infrared spectrum lies just beyond the visible light spectrum as you can see here:

Each portion of the electromagnetic spectrum of light serves its own purpose (gamma rays are used to kill cancer cells, visible light allows for us to see, and infrared allows us to feel heat). We can then harness this penetrating heat in an infrared sauna to put out invisible light that our body absorbs as heat which then raises our core body temperature and produces the amazing health benefits we're already aware of.

There are three levels of infrared, near, mid and far, that each resonate with the subcutaneous fat layer of the human body producing different healing responses. Let's take a look at the

three kinds of infrared light and their benefits in an infrared sauna:

Near Infrared – Healthy cell immunity, wound healing, skin purification and pain relief.

Near infrared wavelengths have a lower micron level and a resulting shorter wavelength which mostly penetrates the epidermis layer of the skin. This makes it a very safe choice for the improvement of cell health, the renewal of skin, tissue growth and wound healing.

A lot of sauna companies use near infrared technology that is similar to the infrared used in handheld cosmetic and LED healing devices. While these LED devices are great when they can be targeted on a specific area of skin—they are ineffective in the sauna environment. Let's take a look at an LED panel that's used by a competitor.

In order for this device to be effective you have to hold it directly over your skin for an extended period of time (minimum 15 minutes to get the blood flowing to the desired area)—not behind a mesh heater pointed at one spot on your back. What happens if you have knee pain? You would have to stand and hold your knee right in front of the LED panel. What's more is that this LED panel is hidden behind a mesh cover which will block nearly all of the rays of light (remember, all light travels in a straight line, including infrared).

The object of an infrared sauna is to heat your body and raise your core temperature—a small 10 watt LED panel is not going to heat your body at all.

What is far better is a near infrared lamp heater which must reach 2,150 °F to produce near infrared waves. You still get all of the amazing health benefits of near infrared (cell health, the renewal of skin, tissue growth and wound healing) while heating your body—which is what an infrared sauna is supposed to do! Furthermore, if you look at companies who only offer near infrared, you'll see they only use near infrared lamps—not LED panels. Why? Because they understand that saunas must be hot!

While near infrared does have great healing benefits, we don't advise using only a near infrared sauna on its own. Near infrared gets too hot to be used solely in an infrared sauna (2,150 °F). Near infrared sauna companies know this and force their sauna users to sit on a small stool and rotate their body at different time intervals essentially rotisserie'ing' themselves. They have to do this because if they surrounded your body on all

four sides it would be far too hot to get any benefit at all from. Not practical or comfortable!

Mid Infrared – Pain relief, improved circulation and weight loss.

Mid infrared wavelengths penetrate deeper into the body than near infrared and are therefore good for increasing circulation and blood flow. When used on a consistent basis, mid infrared sauna therapy has been shown to significantly increase blood flow and reduce muscle and joint pain. Healing can occur faster because mid infrared penetrates deeper into soft tissue where inflammation occurs.

To produce MIR, sauna heaters must be able to maintain a surface temperature of 410 °F. Though major infrared sauna companies may claim to provide mid infrared, the material used in their heaters never actually reaches this temperature. Don't be fooled by fancy marketing and branding. Let the sauna do the talking.

Far Infrared – Weight loss, detoxification, better circulation, reduction in blood pressure, and relaxation.

Because far infrared sauna therapy heats the body directly rather than simply warming the air, it raises the core body temperature and produces a deep, detoxifying sweat at the cellular level, where most toxins reside. Far infrared also aids in blood pressure reduction & weight loss, as you get a great passive aerobic exercise.

The longest infrared wavelength, far infrared, penetrates deepest into the body, reaching adipose tissue (fat cells) where toxins are stored. The range of temperatures that you want for a far infrared heating panel will be from 400 °F to 90 °F. Between these temperatures is where the far infrared band is produced. However, we have found that sitting 2 inches away from a heater that is 400 °F will be far too hot to be comfortable and will not allow you to stay in your sauna for 30 minutes (the object of an infrared sauna is to sit in there longer and prolong your sweat. The longer you sweat the more health benefits you'll get).

Every one of our carbon ("solocarbon," "nanocarbon," etc., they're all the same heater) competitors heaters only ever reach 140 °F on the surface of their heaters which will simply not be hot enough to produce a sweat. For this reason we aim for 200 °F on the surface of our heaters which gives you the longest wavelength at the most beneficial temperature. You will also be able to sit in your sauna for 30 minutes each day while getting the deepest sweat possible.

Why Would You Want a Full Spectrum Infrared Sauna?

For many years Infrared Saunas have been either Far Infrared or Near Infrared. How you used one depended upon your health needs and requirements. Currently an optimal sauna experience combining all three spectrums in one is available, which allows for extended diversity and an incredibly effective sauna experience. This combination of all three spectrums of infrared ensures total health concerns could be addressed and targeted in only one session.

Your skin is the largest organ of your body. The subcutaneous layer of fat stores many toxins and can be found just below the skin's surface. A Full Spectrum Infrared Sauna can penetrate into the tissues up to a depth of two inches. The temperature of a full spectrum infrared sauna is typically between 100 – 150 degrees °F. However, despite operating at a lower temperature, your body will typically perspire far more in a full spectrum infrared sauna due to complete heat penetration.

Full spectrum infrared energy penetrates your body at not just one level, but three.

Normally our bodies pump ten to fourteen pints of blood per minute. Full spectrum infrared sauna rays increase the blood flow up to twenty six pints per minute. When blood flow is increased you have additional oxygenation assisting the body to repair cells and tissues.

The full spectrum infrared sauna also stimulates the sebaceous glands to release harmful toxins such as petrochemicals, mercury and other heavy metals. Toxins that cannot be expelled immediately after entry are stored in our bodies. The full spectrum sauna not only boosts the metabolism but is also extremely effective at aiding the upturn of immune function.

While inside a full spectrum infrared sauna you can fully relax, listen to calming music, meditate, enjoy color light therapy, or perhaps a yoga routine to introduce heat into your practice. After taking a full spectrum sauna session your body's core will be comfortable all day as the detoxing benefits continue

long after your session ends. You'll be relieved of your pain too.

The Ultimate Guide to Far Infrared Sauna Health Benefits

As a former insomnia sufferer, I used to spend evenings afraid. As early as 4 or 5 p.m., a fear would set in—would I be able to sleep that night? This was five years ago, and back then I was stressed, battling a number of health issues, including high blood pressure and being overweight. The single most frustrating issue in my life, however, was my insomnia.

Then I tried my first infrared sauna. I slept like a baby that night (as I still do after every infrared sauna session). In the months and years that have followed, I've continued a steady routine of infrared sauna therapy, and I've learned much more about the different ways these types of saunas benefit the body. I'd like to share some of the broader strokes with you.

How an Infrared Sauna Affects Metabolism

If you know anything about infrared saunas you likely know that the basic goal is to make you sweat by sending infrared light waves—which your body experiences as heat—into the tissues of your body. In order to sweat, your heart must pump at a faster rate, sending more oxygenated blood throughout your body and burning more of the energy you've stored from food and rest. This energy burn is a boosted metabolism, and there are many health benefits that come from putting your body into such a state.

• Weight Loss:

Weight loss is one of the most common reasons people look into saunas, at least initially, and I always assure them that weight loss is a near certainty with regular use of an infrared sauna. The human body's metabolic rate is tied directly to cardiovascular exercise, rising when you perform activities such as swimming, jogging, or playing sports. Even after exercise, a human's metabolism stays boosted for long periods of time. This is also true of using an infrared sauna.

A study done at Binghamton University found that using an infrared sauna three times a week can reduce body fat by about 4 percent over 4 months. For someone who weighs 175 pounds, that comes out to a 7 pound loss, and in this study participants did not change their diet or exercise habits at all. The researchers hypothesized that increasing the body's temperature and metabolism in an infrared sauna elevated human growth hormone production, which helps the body break down fat.

I always tell folks looking to lose weight to be sure to drink plenty of water before and after use of the sauna, which will help speed up their metabolisms even more, and will also

result in their craving water where they once reached for coffee, soda, or alcohol.

- **Injury Recovery:**

Infrared sauna users who have recently suffered certain kinds of injuries may also experience faster recovery times. Joint injury sufferers are generally warned to avoid infrared saunas for at least 48 hours following trauma, but those who have suffered bone fractures or muscle injuries should embrace it. The light absorbed by cells during use of an infrared sauna has been shown to stimulate the metabolism in muscle and bone.

NASA is conducting research right now to determine if the light emitted by LEDs, as well as that of infrared saunas, would help its astronauts and military personnel recover faster from injuries, particularly if they don't have regular exposure to natural sunlight. Their studies have found that light therapy, and speeding up metabolism in bone and muscle, can help to prevent the atrophy to these parts of the body, and can increase the rate of wound healing in an environment without gravity. Basically, NASA is confirming to be true in the laboratory what infrared sauna users have been feeling all along.

Before regular infrared sauna use, you might struggle to lose weight, have to drink extra coffee to wake up due to insomnia, and skip the gym because your sore muscles take days to recuperate.

After regular infrared sauna use, you might see the results of exercise faster, sleep a solid 8 hours with few disturbances, and be able to return to the gym the day after an intense workout.

- **Circulation:**

As I mentioned above, the sweating brought on by the heat from an infrared sauna causes your heart to pump at a faster rate. With more beats per minute, your blood vessels vasodilate, or widen, and more blood moves at a faster rate through the vessels and capillaries beneath the skin. This is an incredible boost to the circulation of blood throughout your body. Since circulation is so important to carry oxygen, infection-fighting cells, nutrients, and hormones around the body, this increased circulation can improve numerous health conditions. It can benefit heart health, blood pressure, cortisol levels, stress, and even cause pain relief.

- **Blood Pressure and Heart Health:**

I had to take medication for high blood pressure before I began using infrared saunas, and I can tell you that suffering from high blood pressure is frightening. If left untreated, high blood pressure can cause heart failure, heart attacks, kidney disease, stroke, and vision loss.

On a more practical level, bad heart health and poor circulation can cause numbness or an irritating cold feeling in your toes and hands, as well as regular headaches. The benefits of

improving your blood circulation and pressure include improved cell growth and organ function, which will have you feeling generally healthier and more energetic.

- **Cortisol Levels and Stress:**

If you've had the heart-pounding experience of speaking in front of a large group of people, you know about fight or flight mode. Stress makes your body go into this mode and release the stress hormone cortisol. But it doesn't take public speaking for these hormones to hurt your body.

Prolonged higher levels of stress cause a constant release of cortisol and adrenaline into your body that is harmful, potentially damaging your immune system, causing you to retain more fat cells, and leaving you prone to depression and insomnia.

I know from experience that lack of sleep can damage your health in so many different ways. It decreases brain function and weakens your immune system. If you're serious about staying in shape, sleep is also the time when your body rebuilds the muscles you worked out on a given day, helping you recover and get stronger more quickly.

Infrared saunas remove your body from fight or flight mode. Inside an infrared sauna, the body relaxes, blood vessels vaso-

dilate, and the user feels as if he has just finished a workout. For me, that feeling of relaxation generally lasts all day and into the night, helping me to sleep so much deeper. My own personal trick is to take a very hot steam shower right after using my sauna, which helps to lull my body into some of the best sleep ever.

- **Pain Relief:**

I know many people whose chronic pain has faded since starting regular infrared sauna use. Infrared saunas relieve pain by producing heat that penetrates joints, muscles, tissues, and nerve endings. They also stimulate the production of additional white blood cells, which alleviate inflammation. Many health experts have come to consider inflammation the underlying cause of many ailments and health conditions. The increased circulation caused by infrared saunas brings healthy, oxygenated blood to whatever area of the body is in need, reducing inflammation.

Before regular infrared sauna use, you might have to take daily medication to keep blood pressure low, sleep fitfully and feel foggy most days, and suffer from painful inflammation-related conditions such as asthma, ulcers and sinusitis.

After regular infrared sauna use, you might get a pat on the back at the doctor's for great blood pressure, wake without an alarm clock, excited to face the day, and smile more, free of nagging pain on a daily basis.

- **Detoxification:**

Detoxification means removing toxic substances from the body, such as drugs, pollutants, preservatives, pesticides, heavy metals, and poison, among other things. Detoxification, I've found, is one of the main reasons people become interested in infrared saunas. Saunas do a good job of detoxification by "vibrating" water inside your cells, helping it move out of the cell and then out of the body—taking toxins with it.

If left inside your body, these toxins can interact with enzymes and confuse cells in a way that keeps them from functioning normally. Documented illnesses stemming from toxicity include cancer, leukemia, liver disease, cardiovascular damage, anemia, nervous system disorders, kidney dysfunction, and reproductive impairment. Besides the potential to improve and prevent these health problems, detoxification can also benefit your appearance by affecting your skin.

- **Skin:**

I didn't start using an infrared sauna to improve my skin, but it's been an amazing bonus. After using an infrared sauna, your skin becomes highly permeable as the blood vessels vasodilate and more blood comes to the surface to cool your body. Millions of sweat glands are activated, and these release and empty to your skin's surface, purging heavy metals and other toxins from your skin layers. This purification proc-

ess leaves your skin looking healthier, smoother, clearer, and softer.

- **Fatigue/Illness Prevention:**

Our everyday environments—like the smog and exhaust fumes I breathe in on my way to work—cause toxins to accumulate in the body that can combine with other disease-causing factors. On a daily basis, toxins can weaken your immune system and leave you more susceptible to illness, as well as fatigue.

If the cells in your body are not reproducing and functioning properly, all manner of routine tasks can become more difficult, leading to exhaustion that you would not otherwise suffer. When detoxified, these cells function better, leading to more energy and a better feeling of health.

Before regular infrared sauna use, you might spend a lot of money on skincare products but still feel that your skin looks too dry or greasy most days, and always take the elevator since the stairs are just too tiring.

After regular infrared sauna use, you might get compliments on your skin with less care products necessary, be surprised in the bathroom mirror by a smooth face, and even take the stairs for a bit of extra exercise whenever possible.

Can an Infrared Sauna Improve Your Life?

Of course it can! Maybe you don't have issues with things like high blood pressure, insomnia, fatigue, high stress, need to lose some weight or detox your body. If that's the case, investing in an infrared sauna could not make more sense for you. If you are starting at a healthy level, there is no way to better protect your health going forward than regular use of an infrared sauna. Additionally, if many of the symptoms previously mentioned seem painfully familiar, there is nothing that will ease that pain more naturally than an infrared sauna.

Take it from me, there's no reason to go on living like a sleepless zombie, struggling to exercise because of nagging body aches. There's a broad range of potential benefits from infrared saunas, but in my experience those benefits all work together. The end result of regular infrared sauna use is the improvement of many separate facets of life that add up to an overall better quality of life by a large margin.

Why Is EMF So Important?

I was a good student in college, but like many students I didn't always find in-depth research thrilling. When I became interested in infrared saunas, though, I found myself doing a great deal of research on my own time about something called electromagnetic fields, or EMF, because it seemed important to know about the health risks.

There is a debate over the dangers of electromagnetic fields, which are emitted by electrical objects like infrared saunas, and I wanted to do my due diligence and review potential risks for sauna users. I learned that the National Institute for Occupational Safety and Health (NIOSH) doesn't consider EMF a proven health hazard, but I also found worrying research suggesting that it might be wise to avoid high-level exposure to EMF, not just in saunas but in our everyday lives.

What Is EMF?

My expertise is in the health and fitness side of the sauna industry, which involves a bit of science but nothing near the in-depth stuff that physicists who study and write about EMF deal with. So I sat down with a physics professor friend to get the basics. He told me that the simplest way to conceive of EMF is to think of invisible energy pulses, and to picture them radiating out from any object that has an electric current flow-

ing inside it. Televisions, computers, microwaves, you name it: if they use electricity, they all emit electromagnetic fields.

My friend is a professor, so of course he wanted to elaborate further, and he added that electromagnetic fields can affect any object—including people—within their field. Most common electrical objects give off such low levels of EMF that they pose no risk to humans. However, larger objects with more electricity running through them, such as power lines, may present a health hazard for people who are exposed to them over long stretches of time.

One last thing: EMF is referred to by some as EMR, or electromagnetic radiation. Now, for our purposes, these are the same thing, so I'll be referring to electromagnetic fields as EMF throughout.

EMF Around the House

As I was describing my conversation with my friend, were you dying to know exactly how strong the EMF emitted by various household objects is? I know I was. My friend, unfortunately, didn't know off the top of his head. That meant I had to do some research of my own.

What I found was that the strength of an electromagnetic field varies massively based on the size and the strength of the item that is emitting it. Later on, I'll talk more about how much EMF infrared saunas tend to emit, but before I do, take a look at this chart of how much EMF is emitted by common household objects. You'll notice that how close you are to an object determines the strength of the EMF that is reaching you.

The Risks of EMF

The next step was to learn about the actual risks associated with EMF. This involved reading quite a few scientific journal articles.

Some of the most significant research conducted on exposure to EMF was reported in an article titled California EMF Risk Evaluation June 2002, which is still cited by many experts in the field today. This article drew from two significant research efforts. The first, which took place in 2001, was ELF Electromagnetic Fields and the Risk of Cancer. Researchers in that study found that high average exposures to EMF could double the risk of leukemia in kids under 15. In a 2002 study, An Evaluation of the Possible Risks from Electric and Magnetic Fields from Power Lines, Internal Wiring, Electrical Occupations and Appliances, researchers found that heavy and prolonged exposure to EMF could lead to increased risk of adult brain cancer, Lou Gehrig's Disease, miscarriage, and, once again, childhood leukemia.

At this point you might be thinking, "But wait, don't a lot of the appliances we use every day give off EMF?" And they do,

but the keywords here are heavy and prolonged. What was discovered in both studies was that the exposure had to be heavy and prolonged. Your microwave giving off one mG of EMF while it cooks a burrito for two minutes, for example, doesn't rank as heavy and prolonged.

However, I don't want to discount the severity of the findings of these studies. My big takeaway was that the less EMF one could be exposed to while using an infrared sauna, the better. As I'd learned earlier though, it is impossible for infrared saunas to emit zero EMF, because they are electrical devices with wiring and currents. Fortunately, there are varieties of infrared sauna that emit barely any EMF at all.

Infrared Saunas and Safe EMF Levels

Infrared saunas are large, they surround you with electrical components, and in order to use them, you must sit inside them for a period of time. If you've followed to this point, you know that that can be a recipe for dangerous exposure to EMF. Thankfully though, there is infrared sauna technology that keeps EMF in the super-low range of .2 to .3 mG. To put that in perspective, the Environmental Protection Agency has previously proposed a safety standard of 3 mG, with anything below 3 mG being deemed as safe. This clear cut numerical guideline makes it easy to shop for a safe infrared sauna: simply ask the sauna salesman for the exact number. Numbers don't lie.

You can never be too cautious, and with that in mind, I suggest that if you're choosing an infrared sauna for your home

that you think about buying a milligauss or Tri-Field EMF meter to test its EMF levels. Hopefully most salesmen out there aren't actively lying about their saunas' EMF levels, but I'd hate for anyone to be deceived. And an EMF meter isn't a bad thing to have around when you're purchasing large electronics.

Choosing a Low-EMF Sauna

The two primary materials used in infrared saunas to produce infrared rays are carbon and ceramic. These materials, when heated, produce infrared rays that raise the body's core temperature, speed up the metabolism, and bring on all the other health benefits of saunas. Carbon heaters are a relatively new invention, and while they have some good qualities, they also generate high EMF levels—up to 80 mG, which is well over the suggested guidelines. Traditional ceramic heaters, on the other hand, have a much lower EMF than carbon heaters, but it's still possible to go lower.

According to testing done by Vitatech Electromagnetics, a third party research group, infrared saunas that have heaters made of a combination of carbon and ceramic emit the smallest amount of EMF, emitting as little as .2 mG. The reason for this is that these heaters use layers of carbon and ceramic compounds on top of each other. When arranged like this, positive and negative charges in the electromagnetic fields produced by each type of material actually cancel each other out, causing the EMF to drop to almost nothing.

In the end, the results of my extensive scientific odyssey were clear. Due to what I learned about EMF, I decided to stick entirely to infrared saunas that use only combination ceramic and carbon heaters, which cut down EMF to miniscule levels. I also encourage everyone I talk to about this to do their own tests with the equipment I mentioned earlier. I know once I'd done them myself, I began to have new peace of mind about EMF risks whenever I was using infrared saunas—and that sort of comfort is invaluable.

How Does My Infrared Sauna Ship?

At Outdoor Infrared Saunas® our goal is to make your infrared sauna buying experience simple, efficient and easy. Our promise is to ensure your total satisfaction. Shipping damage claims are rare, but they do happen. In the event of shipping damage we will promptly replace or fix the damage to your satisfaction. Should you need to return an item to us, please note our Cancellation and Return Policy below:

- **Visible Loss or Damage**

Any external evidence of loss or damage that occurred during transit is considered visible loss or damage and should be noted on the bill of lading and signed by the carrier's agent (driver).

- **Concealed Loss or Damage**

Any damage that occurred during transit that was not visible at the time of delivery is considered concealed loss or damage. Due to rough handling in transit it is possible to damage the contents of a container without damaging the container. If such a situation occurs it is important to contact Outdoor Infrared Sauna Guru immediately for instructions on how to handle the claim. If a claim is to be filed it is important to save the container that the damaged merchandise arrived in. You must open all re-

ceived boxes and inspect products for concealed damage within 7 business days of delivery.

- **Receiving Your Shipment**

When receiving your shipment it is important to do the following:

1. Your shipment may consist of a number of individual boxes placed on a single pallet. The pallet will need to be separated to be unloaded. Your sauna will be shipped with lift gate service with a scheduled appointment for delivery. The driver will move your sauna off of the truck.
2. Check shipment for any visible damage to containers. If the boxes are dented, crushed, scratched, punctured, or cut it is very important to mark this on the bill of lading. If there is a possibility of damage it may be necessary to open the box and mark the actual damage on the bill of lading. It is important to be specific.
3. Boxes that have been dropped may not show obvious signs of external damage. You may want to open these specific boxes before the driver leaves to ensure that the contents do not have concealed damage. The driver may refuse to allow you the time to open specific boxes – if this is the

case, open the boxes as soon as possible, but not later than 7 days after the delivery, to inspect for damage.

4. Refusing a Freight Shipment & Freight Carrier Fees.

If the shipment is damaged and unusable, it may be necessary to refuse part or all of the order. Call our shipping department at 1-855-551-4878 before refusing any shipment.

Please note that any additional freight carrier's fees such as re-delivery are the responsibility of the customer (unless otherwise noted on your order). Liftgate service and delivery notification are standard with all order. If additional services are requested by you and billed to Outdoor Infrared Saunas®, all such fees shall then be re-billed to your credit card. If the freight carrier is unable to deliver the freight due to problems with contacting you or setting up delivery appointments, storage fees may be assessed by the carrier. In such case all storage fees are the responsibility of the customer and shall be billed to your credit card.

Cancellation and Return Policy

1. If an order is cancelled prior to shipment, payment will be fully refunded. Note that we consider your product to be "shipped" when a Bill of Lading and tracking number have been generated for a freight carrier. After products are considered shipped, cancellations are treated as returns as described below.

2. All returns require a Return Merchandise Authorization (RMA) from our Customer Service Department. Please call 1-

855-551-4878 to obtain an RMA. No returns will be accepted or any credit granted without an RMA. To provide any credit or replacement units we must first have a chance to receive and completely inspect your return.

3. Per our "90 Day Risk-Free Return Policy" returns will be accepted within 90 business days of your invoice. Shipping charges on returns shall be the responsibility of the customer and all returned items must be returned in "new" or "like new" condition, in undamaged original manufacturer's packaging and with all original product manuals. After the product is received, inspected and verified to be in "like new" condition, a full refund will be provided on the product purchase price minus any original shipping charges and minus a 15% restock charge. For products sold with a "Free Shipping Promotion" the actual freight cost to originally ship the product will be deducted from your refund. We do not charge any other charges or re-stocking fees for returns.

4. If you refuse an item delivered by a commercial shipping company for any reason other than damaged merchandise or a shipping error by us, the item shall be treated as a return and charges described in Section 3 above shall be applied.

5. All products come with a 3 year manufacturer's warranty which is located in/on their packaging or in their documentation. We offer additional 7 year and lifetime warranties above the manufacturer's warranty.

6. Custom orders, private label or non-standard sauna models are non-refundable.

